

NIKKAT AFRIN

New York, NY | 551-349-1234 | fnunikkatafrin@gmail.com | [linkedin.com/in/nikkat-aftrin](https://www.linkedin.com/in/nikkat-aftrin) | github.com/Nikkat-Afrin | [Tableau Portfolio](#)

SUMMARY

AI Engineer with 2+ years of experience building LLM, RAG, speech-AI, and cloud-based AI workflows that turn student, career, admissions, and logistics data into practical tools. Experienced across Python, FastAPI, Flask, LangChain, ChromaDB, OpenAI APIs, Whisper/faster-Whisper, AWS, Azure, and serverless AI systems.

PROFESSIONAL EXPERIENCE

AI Platform Engineer, Peblink - World Literacy Research Center | New York, NY

Jan 2026 - Present

- Reached **93% pronunciation-detection accuracy** by building a **speech-AI reading assessment workflow** that transcribes and aligns children's oral reading using **Whisper/faster-Whisper**, **FastAPI**, **Python text matching**, and **Levenshtein similarity**.
- Reduced reading-assessment review time from **60+ minutes to under 5 minutes** by deploying a **serverless AWS pipeline** that stores audio/video in **S3**, runs **Lambda/SQS** processing, writes **DynamoDB** results, and supports early **vision-model** work with **Amazon Rekognition**.

AI/Data Analyst (Intern), Peblink - World Literacy Research Center | New York, NY

May 2025 - Jan 2026

- Architected the WLRC global literacy AI/data hub blueprint in **8 weeks** by designing **AWS-based data flows**, **role-based user journeys**, **ingestion pipelines**, and **prototype dashboards** for universities and NGOs; [project featured by YU Katz](#).
- Built a literacy impact benchmarking framework by analyzing **30+ public sources** and comparing programs with **15 - 20% reading-score gains** across reading ability, usability, cultural fit, scalability, and cost using **Python** and **Tableau**.
- Secured a Washington, D.C. meeting with Illinois Congressman Brad Schneider and Peblink leadership by translating **WLRC analytical findings** into AI product requirements that shaped roadmap and partnership decisions.

AI Workflow Engineer (Intern), KAYAAN Inc. (Logistics AI Startup) | New York, NY

Aug 2025 - Dec 2025

- Drove a **24% investor connection rate** and **8% follow-up engagement** by building an **n8n AI-agent workflow** that researched AI and supply-chain investors, generated personalized **LinkedIn messages**, and tracked outreach in **Google Sheets**.
- Enabled leadership to review **15+ operational, engagement, and profitability KPIs** by building a **React analytics dashboard** that surfaced platform usage, driver/load activity, and product-performance metrics.
- Improved release readiness by documenting **20 - 40 QA tickets per sprint**, testing AI-enabled logistics workflows, reproducing bugs, and coordinating fixes with developers before launch.

Graduate Assistant - AI & Data, Yeshiva University, Shevet Glaubach Career Center | New York, NY

Jan 2025 - Dec 2025

- Delivered instant Canvas support for students by co-building **FELIX**, a **RAG chatbot** that answers Canvas navigation, course-material, and career-resource questions using **LangChain**, **ChromaDB**, **OpenAI GPT-4/3.5**, **Flask**, and **Azure**.
- Eliminated **40+ staff hours per week** of manual reporting and improved advising productivity by **12%** by automating **Symlicity CSM** ingestion through **Azure Functions**, **Blob Storage**, **PostgreSQL**, and **Databricks**.

Graduate Assistant - AI & Admissions, Yeshiva University, Katz School of Science & Health | New York, NY

Jun 2024 - Aug 2025

- Accelerated student-response support by building an **admissions ChatGPT assistant** that helped GAs answer applicant questions and draft emails using **OpenAI prompts**, reusable response templates, and admissions knowledge content.
- Improved admissions decision-making by building **Power BI reports** that helped leadership track applicant funnel stages, outreach activity, application status, and enrollment trends across **Recruit Ellucian** and **HubSpot data**.

PUBLICATIONS

SigmaCam: Exact Decision Boundary Extraction for DNNs (First Author) - Published, [IJCNN 2025 \(IEEE Xplore\)](#)

Mar 2025

- Improved AI model interpretability by introducing **SigmaCam**, a **PyTorch-accelerated algorithm** that extracts exact decision boundaries for smooth neural-network activations like **Sigmoid** and **SiLU**, reducing runtime by **~93%** while maintaining **~99.8% accuracy**.

SigmaCam++: Generalizing Exact Decision Boundary Extraction for DNNs with Smooth Nonlinearities (First Author) - Submitted, IEEE TNNLS

2026

PROJECTS

Azure Real-Time Voice Conversational AI System | [GitHub](#)

- Enabled real-time, hands-free AI conversations by building a voice assistant that captures speech, transcribes user input, retrieves relevant context, generates responses, and speaks back using **Azure Speech Services**, **Azure OpenAI**, **Azure AI Search**, **FastAPI/WebSockets**, and **Azure Container Apps**.

EDUCATION

Yeshiva University, Katz School of Science & Health - M.S. Data Analytics & Visualization | GPA: 3.92 | New York, NY Jan 2024 - Jan 2026

Relevant course work:

Andhra University (Gold Medalist) - Bachelor of Computer Applications | GPA: 4.0/4.0 | India

Jun 2019 - Jul 2023

TECHNICAL SKILLS

AI/ML & GenAI: LLMs, RAG, LangChain, ChromaDB, OpenAI APIs, GPT-4/3.5, Prompt Engineering, Whisper, faster-Whisper, Amazon Transcribe, Amazon Rekognition, Speech-to-Text, Transcript Alignment, Pronunciation Scoring, Vision Models, Hugging Face, PyTorch **Cloud, Backend & MLOps:** AWS S3, Lambda, SQS, API Gateway, Cognito, DynamoDB, Athena, QuickSight, Azure Functions, Azure Web Apps, Terraform, Docker, FastAPI, Flask, REST APIs, Git/GitHub, GitHub Actions, Serverless Architecture

Data, Analytics & Programming: Python, SQL, Pandas, NumPy, scikit-learn, Power BI, Tableau, Excel, PostgreSQL, MS SQL Server, MySQL, Databricks, Apache Spark, PySpark, Data Modeling, KPI Dashboards, QA Testing

AWARDS & RECOGNITION

- Best Research Project Award** - Yeshiva University, Spring 2025 CSE Research Forum - Won for first-author SigmaCam research advancing AI model transparency through exact decision boundary extraction for neural networks. [Feature Article](#)
- Innovation Lab Challenge Finalist** - Yeshiva University, 2025 - Selected as **1 of 7 finalist teams from 30 original teams / 84 registrants** to pitch CloselQ, an AI-powered sales prospecting tool. [LinkedIn Post](#)